

**[San Jose State University Special AI Lecture Series IV -
Philosophy, Ethics & Consciousness]
AI's Philosophical Frontiers - Consciousness, Knowledge,
Belief, Ethics, and Limits of Machine Reasoning**

Sunghee Yun

Co-Founder & CTO @ Erudio Bio, Inc.

Co-Founder & CEO @ Erudio Bio Korea, Inc.

Leader of Silicon Valley Privacy-Preserving AI Forum (K-PAI)

CGO / Global Managing Partner @ LULUMEDIC

Global Leadership Initiative Fellow @ Salzburg Global Seminar

Visiting Professor & Advisory Professor @ Sogang Univ. & DGIST

About Speaker

- *Co-Founder & CTO @ Erudio Bio, Inc., San Jose & Novato, CA, USA* 2023 ~
- *Co-Founder & CEO @ Erudio Bio Korea, Inc., Korea* 2025 ~
- *Leader of Silicon Valley Privacy-Preserving AI Forum (K-PAI), CA, USA* 2024 ~
- *CGO / Global Managing Partner @ LULUMEDIC, Seoul, Korea* 2025 ~
- *KFAS-Salzburg Global Leadership Fellow @ Salzburg Global Seminar, Austria* 2024 ~
- *Adjunct Professor, EE Department @ Sogang University, Seoul, Korea* 2020 ~
- *Advisory Professor, EECS Department @ DGIST, Korea* 2020 ~
- *AI-Korean Medicine Integration Initiative Task Force Member @ The Association of Korean Medicine, Seoul, Korea* 2025 ~
- *Director of AI Semiconductor @ K-BioX, CA, USA* 2025 ~
- Global Advisory Board Member @ Innovative Future Brain-Inspired Intelligence System Semiconductor of Sogang University, Korea 2020 ~
- Technology Consultant @ Gerson Lehrman Group (GLG), NY, USA 2022 ~
- Chief Business Development Officer @ WeStory.ai, Cupertino, CA, USA 2025 ~
- Advisor @ CryptoLab, Inc., Seoul, Korea 2025 ~

- Co-Founder & CTO / Head of Global R&D / Chief Applied Scientist / Senior Fellow @ Gauss Labs, Inc., Palo Alto, CA, USA 2020 ~ 2023
- Senior Applied Scientist @ Amazon.com, Inc., Vancouver, BC, Canada 2017 ~ 2020
- Principal Engineer @ Software R&D Center, Samsung Electronics 2016 ~ 2017
- Principal Engineer @ Strategic Marketing & Sales, Memory Business 2015 ~ 2016
- Principal Engineer @ DT Team, DRAM Development, Samsung 2012 ~ 2015
- Senior Engineer @ CAE Team, Memory Business, Samsung, Korea 2005 ~ 2012
- PhD - Electrical Engineering @ Stanford University, CA, USA 2001 ~ 2004
- Development Engineer @ Voyan, Santa Clara, CA, USA 2000 ~ 2001
- MS - Electrical Engineering @ Stanford University, CA, USA 1998 ~ 1999
- BS - Electrical & Computer Engineering @ Seoul National University 1994 ~ 1998

Highlight of Career Journey

- BS in Electrical Engineering (EE) @ Seoul National University
- MS & PhD in Electronics Engineering (EE) @ Stanford University
 - *Convex Optimization - Theory, Algorithms & Software*
 - advisor - *Prof. Stephen P. Boyd*
- Principal Engineer @ Samsung Semiconductor, Inc.
 - *AI & Convex Optimization*
 - collaboration with *DRAM/NAND Design/Manufacturing/Test Teams*
- Senior Applied Scientist @ Amazon.com, Inc.
 - *e-Commerce AIs* - anomaly detection, deep RL, and recommender system
 - *Jeff Bezos's project - drove \$200M* in sales via Amazon Mobile Shopping App
- *Co-Founder & CTO / Global R&D Head & Chief Applied Scientist* @ Gauss Labs, Inc.
- *Co-Founder & CTO* @ Erudio Bio, Inc.
- *Co-Founder & CEO* @ Erudio Bio Korea, Inc.

Unpacking AI

- Some Important Questions around AI - 5
 - Why Human level AI?
 - Why sudden leap in LLM performance?
 - Biases
 - AI ethics
 - Consciousness
 - Can AI think, reason, believe, or know something?
 - Risk of anthropomorphization
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Some Important Questions around AI

Some important questions around AI

- why human-level AI?
- what lies in very core of DL architecture? what makes it work amazingly well?
- biases that can hurt judgement, decision making, social good?
- AI ethics & legal issues
- consciousness
- utopia vs dystopia
- knowledge, belief, reasoning
- risk of anthropomorphization

Human-level AI?

Why human-level in the first place?

- lots of times, when we measure AI performance, we say
 - how can we achieve human-level performance, *e.g.*, CV models?
- why human-level?
 - are all human traits desirable? are humans flawless?
 - aren't humans still evolving?
- advantage of AI over humans
 - *e.g.*, self-driving cars can use extra eyes, GPS, computer network
 - *e.g.*, recommendation system runs for hundreds of millions of people overnight
 - AI is available 24 / 7 while humans cannot
 - . . . critical advantages for medical assistance, emergency handling
 - AI does not make more mistakes because task is repetitive and tedious
 - AI does not request salary raise or go on strike

What makes DL so successful?

Factors contributing to astonishing success of DL

- analysis based on speaker's mathematical, numerical algorithmic & statistical perspectives considering hardware innovations

30% universal approximation theorem? - (partially) yes! but that's not all

- function space of neural network is *dense* (math theory), *i.e.*, for every $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$, exists $\langle f_n \rangle$ such that $\lim_{n \rightarrow \infty} f_n = f$

25% architectures/algorithms tailored for each class of applications, *e.g.*, CNN, RNN, Transformer, NeRF, diffusion, GAN, VAE, . . .

20% data labeling - expensive, data availability - unlimited web text corpus

15% computation power/parallelism - AI accelerators, *e.g.*, GPU, TPU & NPU

10% rest - Python, open source software, cloud computing, MLOps, . . .

Sudden leap in LLM performance

Probability inferred sequence is correct

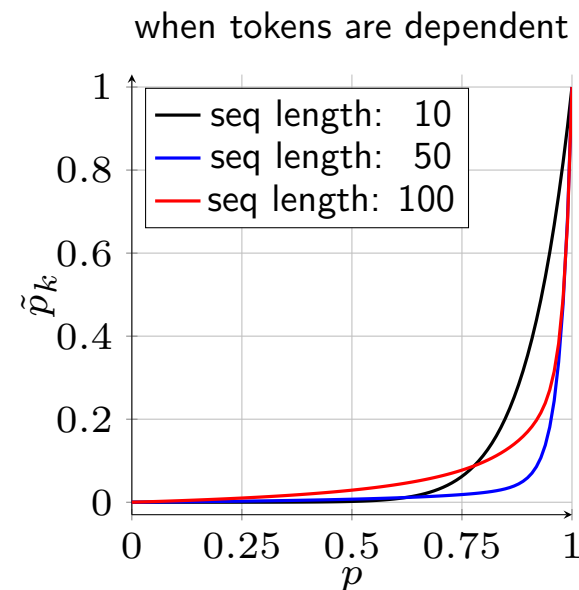
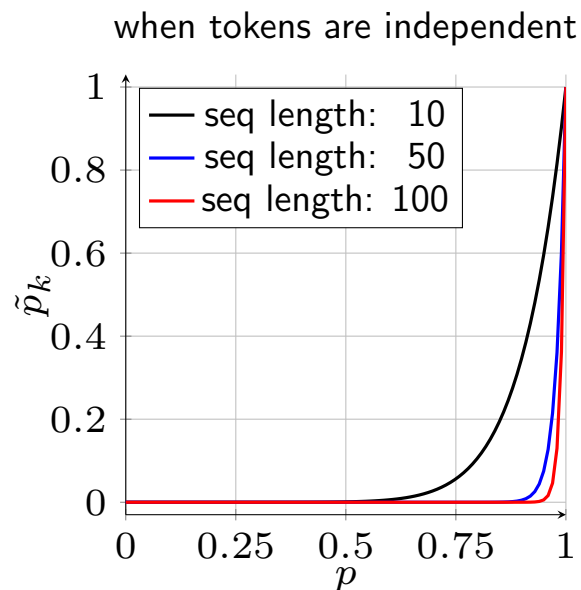
- assume
 - t_i - i th token
 - p_i - probability that t_i is correct
 - ρ_i - correlation coefficient between t_{i-1} & t_i
 - $\tilde{p}_k$ - probability that $(t_1, \dots, t_k)$ are correct
- recursion

$$\rho_i = \frac{\tilde{p}_i - \tilde{p}_{i-1}p_i}{\sqrt{\tilde{p}_{i-1}(1 - \tilde{p}_{i-1})p_i(1 - p_i)}}$$

$$\Leftrightarrow \tilde{p}_i = \tilde{p}_{i-1}p_i + \rho_i \sqrt{\tilde{p}_{i-1}(1 - \tilde{p}_{i-1})p_i(1 - p_i)}$$

Dramatic improvement of LLM near saturation

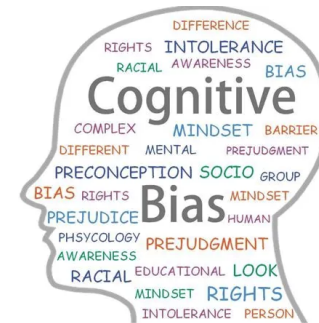
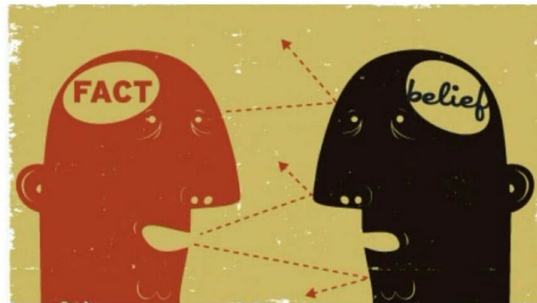
- do simulations for both independent & dependent cases
 - assume p_i are same for all i
- (for both cases) sequence inference improves dramatically as p approaches 1
- this explains *why we have observed sudden dramatic performance improvement of certain seq2seq learning technologies*, e.g., LLM



Biases

Cognitive biases attributed to humans

- cognitive biases [[Kah11](#)]
 - confirmation bias, availability bias
 - hindsight bias, confidence bias, optimistic bias
 - anchoring bias, halo effect, framing effect, outcome bias
 - belief bias, negativity bias, false consensus



Biases of LLMs

- LLMs subject to
 - availability bias - biased by imbalancedly available information
 - LLM trained by imbalanced # articles for specific topics
 - belief bias - derive conclusion not by reasoning, but by what it saw
 - LLM easily inferencing what it saw, *i.e.*, data it trained on
 - halo effect - overemphasize on what prestigious figures say
 - LLM trained by imbalanced # reports about prestigious figures
- similar facts true for other types of ML models,
 - *e.g.*, video caption, text summarization, sentiment analysis
- cognitive biases only human represent
 - confirmation bias, hindsight bias, confidence bias, optimistic bias, anchoring bias, negativity bias, framing effect

AI Ethics

Ethical issues related to AI

- AI can be exploited by those who have bad intention to
 - manipulate / deceive people - using manipulated data corpus for training
 - *e.g.*, spread false facts
 - induce unfair social resource allocation
 - *e.g.*, medical insurance, taxation
 - exploit advantageous social and economic power
 - *e.g.*, unfair wealth allocation, mislead public opinion
- AI for Good - advocated by Andrew Ng
 - *e.g.*, public health, climate change, disaster management
- should scientists and engineers be morally & politically conscious?
 - *e.g.*, Manhattan project

AI related Legal Issues

Legal issues with ethical consideration

- scenario 1 - full self-driving algorithm causes traffic accident killing people
 - who is responsible? - car maker, algorithm developer, driver, algorithm itself?
- scenario 2 - self-driving cars kill less people than human drivers
 - *e.g.*, human drivers kill 1.5 people for 100,000 miles & self-driving cars kill 0.2 people for 100,000 miles
 - how should law makers make regulations?
 - utilitarian & humanitarian perspectives
- scenario 3 - someone is not happy with their data being used for training
 - “The Times sues OpenAI and Microsoft over AI use of copyrighted work” (Dec-2023)
 - “Newspaper publishers in California, Colorado, Illinois, Florida, Minnesota and New York said Microsoft and OpenAI used millions of articles without payment or permission to develop ChatGPT and other products” (Apr-2024)

Consciousness

Consciousness

- what is consciousness, anyway?
 - recognizes itself as independent, autonomous, valuable entity?
 - recognizes itself as living being, unchangeable entity?
- no agreed definition on consciousness exists yet
... and will be so forever
- does it have anything to do with the fact that humans are biologically living being?
- is SKYNET ever plausible?
 - can AI have *desire* to survive (or save earth)?



Utopia vs Dystopia

Utopia vs dystopia



- not important questions (at all) *I think . . .*
- what we should focus on is *not* the possibilities of doomday or Judgment Day, but rather
 - our limits on controlling unintended impacts of AI
 - *misuse* by (greedy, immoral, and unethical) people possessing social, economic & political power
 - *social good and welfare impaired* by either exploiting AI or ignorance of (inner workings of) AI
- should concern
 - choice or balance among utilitarianism, humanitarianism & values
 - amend or improve laws/regulations
 - ethical issues caused by AI

Knowledge, Belief, and Reasoning

Does AI (LLM) have knowledge or belief? Can it reason?

**What categories of questions do they belong to?
engineering, scientific, philosophical, cognitive scientific, . . . ?**

LLMs . . .

- LLM is very different sort of animal . . . except that it is *not* an animal!
- *unreasonable* effectiveness of data [HNF09]
 - *performance scales with size of training data*
 - *qualitative leaps* in capability as models scale
 - tasks demanding human intelligence *reduced to next token prediction*
- focus on third surprise

conditional probability model looks like human with intelligence

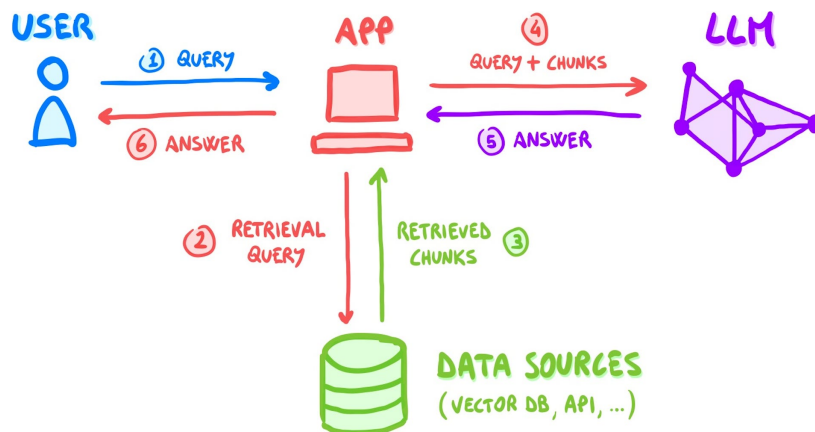
- making vulnerable to anthropomorphism
- examine it by throwing questions such as
 - “*does LLM have knowledge and belief?*”
 - “*can it reason?*”

What LLM really does!

- given prompt “the first person to walk on the Moon was”, LLM responds with “Neil Armstrong” . . . strictly speaking
 - it’s *not* being asked *who* was the first person to walk on the Moon
 - what are being *really* asked is *“given statistical distribution of words in vast public corpus of text, what words are most likely to follow ‘The first person to walk on the Moon was’?”*
- given prompt “after ring was destroyed, Frodo Baggins returned to”, LLM responds with “the Shire”
 - on one level, it seems fair to say, you might be testing LLM’s knowledge of fictional world of Tolkien’s novels
 - what are being *really* asked is *“given statistical distribution of words in vast public corpus of text, what words are most likely to follow ‘After the ring was destroyed, Frodo Baggins returned to’?”*

LLMs vs systems in which they are embedded

- crucial to distinguish between the two (for philosophical clarity)
 - LLM (bare-bones model) - highly specific & well-defined function, which is *conditional probability estimator*
 - systems in which LLMs are embedded, *e.g.*, for question-answering, news article summarization, screenplays generation, language translation



How ChatBot works?

- conversational AI agent does *in-context learning* or *few-shot prompting*
- for example,
 - when the user enters
who is the first person to walk on the Moon?
 - ChatBot, LLM-embedded system, feeds the following to LLM
User, a human, and BOT, a clever and knowledgeable AI agent.
User: what is 2+2?
BOT: the answer is 4.
User: where was Albert Einstein born?
BOT: he was born in Germany.
User: who is the first person to walk on the Moon?
BOT:

Knowledge, belief & reasoning around LLM

- *not* easy topic to discuss, or even impossible because
 - we *do not have agreed definition* of these terms especially in context of being asked questions like

does LLM have belief?

or

do humans have knowledge?

- let us discuss them in two different perspectives
 - laymen's perspectives
 - cognitive scientific & philosophical perspectives

Laymen's perspectives on knowledge, belief & reasoning

- does (good) LLM have knowledge?
 - Grandmother: looks like it cuz when instructed *“explaining big bang”*, it says
“ The Big Bang theory is prevailing cosmological model that explains the origin and evolution of the universe. . . . 13.8 billion years ago . . . ”
- does it have belief?
 - Grandmother: I don't think so, *e.g.*, it does not believe in God!
- can it reason?
 - Grandmother: seems like it! *e.g.*, when asked *“Sunghee is a superset of Alice and Beth is a superset of Sunghee. is Beth a superset of Alice?”*, it says
“ Yes, based on information provided, if Sunghee is a superset of Alice and Beth is a superset of Sunghee, then Beth is indeed a superset of Alice . . . ”
- can it reason to prove theorem whose inferential structure is more complicated?
 - Grandmother: I'm not sure – actually, I don't know what you're talking about!

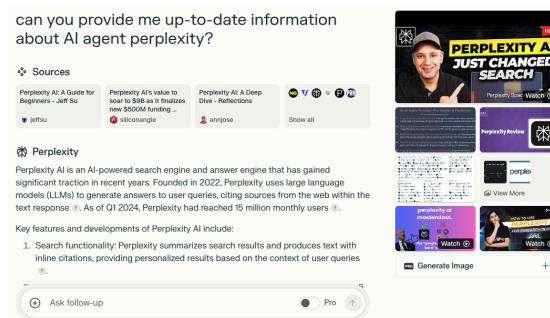
Knowledge

- could argue LLM “knows” which words follow which other words with high probability
- but, only *in context of capacity to distinguish truth from falsehood* can we legitimately speak of “knowledge”!
- LLM(-embedded BOT)
 - can be said to “*encode*”, “*store*”, or “*contain*” knowledge
 - lacks means to use words “true” & “false” in all ways & in all contexts because . . .
 - *does not inhabit the world* we human language-users share!



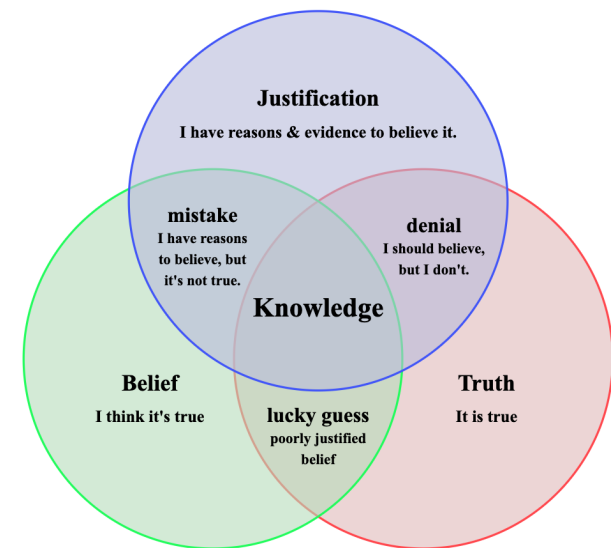
Belief

- nothing can count as *belief about the world* we share unless
 - is against backdrop of “*ability to update beliefs appropriately in light of evidence from that world*” - (again) essential capacity to distinguish truth from falsehood
- change taking place in humans when acquiring or updating belief is
 - reflection of their nature as language-using animals inhabiting shared world with community of language-users
- then, *what if LLM-embedded system updates LLM with outside world information?*
 - even so, when interacting with AI systems based on LLMs, these grounds are *absent!*



Knowledge in philosophical and cognitive scientific sense

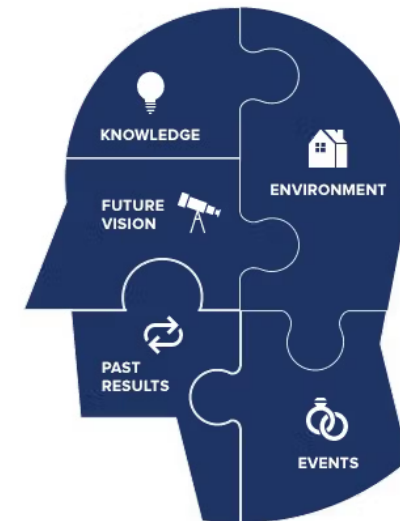
- does LLM have knowledge?
 - Sunghee: *I don't think so!*
- why?
 - we say we have “knowledge” when
“we do so against ground of various human capacities that we all take for granted when we engage in everyday conversation with each other.”
 - when asked *“who is Tom Cruise's mother?”*, it says *“Tom Cruise's mother is Mary Lee Pfeiffer.”*
However, this is nothing but
“guessing” by conditional probability model the most likely words following “Tom Cruise's mother is.”
 - so *we cannot say it really knows the fact!*



Belief in philosophical and cognitive scientific sense

- for the discussion
 - do *not* concern any specific belief
 - but concern *prerequisites for ascribing any beliefs to AI system*
- so does it have belief?
 - nothing can count as belief about the world we share unless
 - it is against ground of the ability to update beliefs appropriately in light of evidence from that world, essential aspect of the capacity to distinguish truth from falsehood*
 - LLM does not have this ground, essential consideration when deciding whether it *really* had beliefs.
- Sunghee: so *no, LLM cannot have belief!*

WHERE DO YOUR BELIEFS COME FROM?



Reasoning in philosophical and cognitive scientific sense

- note reasoning is *content neutral*
 - e.g., following logic is perfect regardless of truth of premises
 - hence, no access to outside world does *not* disqualify
- when asked “*if humans are immortal, would Socrates have survived today?*”, LLM says “*. . . it’s logical to conclude that Socrates would likely still be alive today. . . .*”
- however, remember, once again, what we just asked it to do is *not* “deductive inference”
given the statistical distribution of words in public corpus, what words are likely to follow the sequence, “humans are immortal and Socrates is human therefore.”
- Sunghee: so *no, LLM cannot reason, either!*
- but, LLM
 - pretends to reason, and from which capabilities, we can benefit!
 - also, can *mimic even multi-step reasoning whose inferencing structure is complicated* using *chain-of-thoughts prompting*, i.e., *in-context learning* or *few-shot prompting*

Simple example showing LLM not possessing knowledge

- User

"Who is Tom Cruise's mother?"



- LLM(-embedded question-answering system) (as of Jan 2022)

"Tom Cruise's mother is Mary Lee Pfeiffer. She was born Mary Lee South. . . . Information about his family, including his parents, has been publicly available, . . . "

- User

"Who is Mary Lee Pfeiffer's son?"

- LLM(-embedded question-answering system) (as of Jan 2022)

"As of my last knowledge update in January 2022, I don't have specific information about Mary Lee Pfeiffer or her family, including her son. . . . "

Risk of anthropomorphization

- unfortunately, contemporary LLMs are *too powerful, too versatile, and too useful for most people to accept (after understanding) previous arguments!*
- maybe, o.k. for laymen to (mistakenly) anthropomorphize LLM(-embedded systems)
- however, *imperative for (important, smart, and responsible) AI researchers, scientists, engineers & practitioners* to have rigorous understanding in these aspects especially when
 - advise and be consulted by law makers, policy makers, journalists, and various stakeholders responsible for *critical business decisions (in private sectors) and public policies (in public sectors)*
 - collaborate with or/and help professionals in liberal arts, such as *philosophy, ethics, law, religion, literature, history, music, cultural studies, psychology, sociology, anthropology, political science, economics, archaeology, linguistics, media studies, natural sciences, fine arts, . . .*
 - to address negative societal and economic impacts

Moral

- AI shows incredible utility and commercial potentials, hence should
 - make informed decisions about trustworthiness and safety
 - avoid ascribing capacities they lack
 - *take best utilization of remarkable capabilities of AI*
- today's AI so powerful, so (seemingly) convincingly intelligent
 - obfuscate mechanism
 - actively encourage *anthropomorphism* with philosophically loaded words like *“believe”* and *“think”*
 - easily mislead people about character and capabilities of AI
- matters not only to scientists, engineers, developers, and entrepreneurs, but also
 - *general public, law & policy makers, journalists, . . .*

Selected References & Sources

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- CEOs, CTOs, CFOs, COOs, CMOs & CCOs @ startup companies in Silicon Valley
- VCs on Sand Hill Road - Palo Alto, Menlo Park, Woodside in California, USA

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- [HNF09] Alon Halevy, Peter Norvig, and Nanediri Fernando. The unreasonable effectiveness of data. *Intelligent Systems, IEEE*, 24:8 – 12, 05 2009.
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Thank You